CISS240: Introduction to Programming
Quiz q1203

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This is a closed-book, no compiler, 5 minute quiz.

Q1. Write a program that simulates the rolling of two dice until the sum of the two dice is 10. The program prints the dice values of each throw. You must use a while-loop. The output of a sample execute is given below (notice that the last output has two integer values adding up to 10.) Here's a test run:

```
1 5
2 3
5 6
6 4
```

Answer:

```
#include <iostream>
#include <ctime>

int main()
{
    srand((unsigned int) time(NULL));

    while (true)
    {
        // generate a die roll and store it in die0
        int die0 = rand() % 6 + 1;
        // generate a die roll and store it in die1
        int die1 = rand() % 6 + 1;
        // print die0, die1
        std::cout << die0 << ' ' << die1 << '\n';
        // if sum of die0 and die1 is 10, break
        if(die0 + die1 == 10)
        {
             break;
        }
    }

    return 0;
}
```

Q2. Write a program that gets a positive integer n from the user and continually compute a sequence of integers starting with n. Here's how the sequence is computed.

If a number x in the sequence is even, the *next* number after x is $x/2$. If a number x in the sequence is odd, the *next* number is $3x + 1$. Once the sequence reaches 1, the sequence stops. For instance suppose the integer entered by the user is 7. Then the sequence is $7, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1$. (7 is odd, so the next number is $3 \cdot 7 = 1 = 22$. 22 is even, so the next number is $22/2 = 11$. Etc.) Here's a test run:

```
7
7 22 11 34 17 52 26 13 40 20 10 5 16 8 4 2 1
```

Answer:

```
#include <iostream>

int main()
{
    int n;
    std::cin >> n;

    while (n != 1)
    {
        std::cout << n << " ";

        if (n % 2 == 0) // If the number is even
        {
            n /= 2;
        }
        else // If the number is odd
        {
            n = 3 * n + 1;
        }
    }

    std::cout << n << std::endl;
    return 0;
}
```

(The above procedure always produce a sequence that reaches 1. But the reason is not known and is a famous unsolved problem called Collatz Conjecture.)

Instructions

In the file `thispreamble.tex` look for

```
\renewcommand\AUTHOR{}
```

and enter your email address:

```
\renewcommand\AUTHOR{jdoe5@cougars.ccis.edu}
```

(This is not really necessary since alex will change that for you when you execute `make`.) In your bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

Enter your answers in `main.tex`. In the bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

For each question, you'll see boxes for you to fill. For small boxes, if you see

```
1 + 1 = \answerbox{}.
```

you do this:

```
1 + 1 = \answerbox{2}.
```

`answerbox` will also appear in "true/false" and "multiple-choice" questions.

For longer answers that need typewriter font, if you see

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
\end{answercode}
```

you do this:

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
int x;
\end{answercode}
```

`answercode` will appear in questions asking for code, algorithm, and program output. In this case, indentation and spacing is significant. For program output, I do look at spaces and newlines.

For long answers (not in typewriter font) if you see

```
What is the color of the sky?
\begin{answerlong}
\end{answerlong}
```

you can write

```
What is the color of the sky?
\begin{answerlong}
The color of the sky is blue.
\end{answerlong}
```

A question that begins with "T or F or M" requires you to identify whether it is true or false, or meaningless. "Meaningless" means something's wrong with the question and it is not well-defined. Something like "$1 + 2 = 4$" is either true or false (of course it's false). Something like "$1+_2 = 4$?" does not make sense.

When writing results of computations, make sure it's simplified. For instance write 2 instead of $1 + 1$.

Higher level classes.

For students beyond 245: You can put LaTeX commands in `answerlong`.

More examples of meaningless statements: Questions such as "Is $42 = 1+_2$ true or false?" or "Is $42 = \{2\}^{\{3\}}$ true or false?" does not make sense. "Is $P(42) = \{42\}$ true or false?" is meaningless because $P(X)$ is only defined if X is a set. For "Is $1 + 2 + 3$ true or false?", "$1 + 2 + 3$" is well–defined but as a "numerical expression", not as a "proposition", i.e., it cannot be true or false. Therefore "Is $1 + 2 + 3$ true or false?" is also not a well-defined question.

More examples of simplification: When you write down sets, if the answer is $\{1\}$, do not write $\{1, 1\}$. And when the values can be ordered, write the elements of the set in ascending order. When writing polynomials, begin with the highest degree term.

When writing a counterexample, always write the simplest.